REPORT 1381

A STUDY OF THE MOTION AND AERODYNAMIC HEATING OF BALLISTIC MISSILES ENTERING THE EARTH'S ATMOSPHERE AT HIGH SUPERSONIC SPEEDS [1]

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SUMMARY

A simplified analysis is made of the velocity and deceleration history of ballistic missiles entering the earth's atmosphere at high supersonic speeds. It is found that, in general, the gravity force is negligible compared to the aerodynamic drag force and, hence, that the trajectory is essentially a straight line. A constant drag coefficient and an exponential variation of density with altitude are assumed and generalized curves for the variation of missile speed and deceleration with altitude are obtained. A curious finding is that the maximum deceleration is independent of physical characteristics of a missile (e. g., mass, size, and drag coefficient) and is determined only by entry speed and flight-path angle, provided this deceleration occurs before impact.

The results of the motion analysis are employed to determine means available to the designer for minimizing aerodynamic heating. Emphasis is placed upon the convective-heating problem including not only the total heat transfer but also the maximum average and local rates of heat transfer per unit area. It is found that if a missile is so heavy as to be retarded only slightly by aerodynamic drag, irrespective of the magnitude of the drag force, then convective heating is minimized by minimizing the total shear force acting on the body. This condition is achieved by employing shapes with a low pressure drag. On the other hand, if a missile is so light as to be decelerated to relatively low speeds, even if acted upon by low drag forces, then convective heating is minimized by employing shapes with a high pressure drag, thereby maximizing the amount of heat delivered to the atmosphere and minimizing the amount delivered to the body in the deceleration process. Blunt shapes appear superior to slender shapes from the standpoint of having lower maximum convective heat-transfer rates in the region of the nose. The maximum average heat-transfer rate per unit area can be reduced by employing either slender or blunt shapes rather than shapes of intermediate slenderness. Generally, the blunt shape with high pressure drag would appear to offer considerable promise of minimizing the heat transfer to missiles of the sizes, weights, and speeds of usual interest.

INTRODUCTION

For long-range ballistic trajectories one of the most difficult phases of flight the designer must cope with is the reentry into the earth's atmosphere, wherein the aerodynamic heating associated with the high flight speeds is intense. The air temperature in the boundary layer may reach values in the tens of thousands of degrees Fahrenheit which, combined with the high surface shear, promotes very great convective heat transfer to the surface. Heat-absorbent material must therefore be provided to prevent destruction of the essential elements of the missile. It is characteristic of long-range rockets that for every pound of material which is carried to "burn-out," many pounds of fuel are required in the booster to obtain the flight range. It is clear, therefore, that the amount of material added for protection from excessive aerodynamic heating must be minimized in order to keep the take-off weight to a practicable value. The importance of reducing the heat transferred to the missile to the least amount is thus evident.

For missiles designed to absorb the heat within the solid surface of the missile shell, a factor which may be important, in addition to the total amount of heat transferred, is the rate at which it is transferred since there is a maximum rate at which the surface material can safely conduct the heat within itself. An excessively high time rate of heat input may promote such large temperature differences as to cause spalling of the surface, and thus result in loss of valuable heat-absorbent material, or even structural failure as a result of stresses induced by the temperature gradients.

For missiles designed to absorb the heat with liquid coolants (e. g., by transpiration cooling where the surface heat-transfer rate is high, or by circulating liquid coolants within the shell where the surface heat-transfer rate is lower), the time rate of heat transfer is similarly of interest since it determines the required liquid pumping rate.

These heating problems, of course, have been given considerable study in connection with the design of particular missiles, but these studies are very detailed in scope. There has been need for a generalized heating analysis intended to show in the broad sense the means available for minimizing the heating problems. Wagner, reference 1, made a step toward satisfying this need by developing a laudably simple motion analysis. This analysis was not generalized, however, since it was his purpose to study the motion and heating of a particular missile.

It is the purpose of this report to simplify and generalize the analysis of the heating problem in order that the salient features of this problem will be made clear so that successful solutions of the problem will suggest themselves.

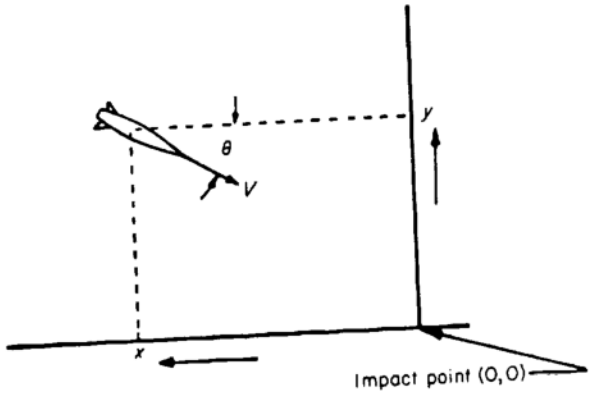
A motion analysis, having the basic character of Wagner's approach, precedes the heating analysis. The generalized results of this analysis are of considerable interest in themselves and, accordingly, are treated in detail.

[1] Supersedes NACA Technical Note 4047 by H. Julian Allen and A. J. Eggers, Jr., 1957.

ANALYSIS

MOTION OF THE BODY

Consider a body of mass m entering the atmosphere from great height. If, at any altitude y, the speed is V and the angle of approach is θ to the horizontal (see sketch), the



parametric equations of motion can be written [2]

$$\left.\begin{aligned}\frac{d^2y}{dt^2}&=-g+\frac{C_D\rho V^2A}{2m}\sin\theta\\ \frac{d^2x}{dt^2}&=\frac{C_D\rho V^2A}{2m}\cos\theta\end{aligned}\right\} \tag{1}$$

where

C_D drag coefficient, dimensionless
V speed, ft/sec
A reference area for drag evaluation, sq ft
m mass of the body, slugs
ρ mass density of the air, slugs/ft^3
g acceleration of gravity, ft/sec^2
x,y horizontal and vertical distance from the point of impact with the earth, ft
θ angle between the flight path and the horizontal, deg

(See Appendix A for complete list of symbols.)

In general, the drag coefficient varies with Mach number and Reynolds number, while the density and, to a very minor extent, the acceleration of gravity vary with altitude. Hence it is clear that exact solution of these equations is formidable. Let us first, then, consider the following simplified case:

1. The body descends vertically.
2. The drag coefficient is constant.[3]
3. The acceleration of gravity is constant.[4]
4. The density as a function of altitude is given by the relation

$$\rho=\rho_o e^{-\beta y} \tag{2}$$

where ρ_o and β are constants. This relation is consistent with the assumption of an isothermal atmosphere.

Equations (1) then reduce to the single equation

$$\frac{d^2y}{dt^2}=-\frac{dV}{dt}=-g+\frac{C_D\rho_oA}{2m}e^{-\beta y}V^2 \tag{3}$$

Noting that

$$\frac{dV}{dt}=-V\frac{dV}{dy}$$

we let

$$Z=V^2$$

and equation (3) becomes the linear differential equation

$$\frac{dZ}{dy}-\frac{C_D\rho_oA}{m}e^{-\beta y}Z+2g=0 \tag{4}$$

which has the well-known solution

$$Z=e^{\int\frac{C_D\rho_oA}{m}e^{-\beta y}dy}\left(-2g\int e^{-\int\frac{C_D\rho_oA}{m}e^{-\beta y}dy}dy+const.\right)$$

Performing the integrations, we obtain as the solution of this relation

$$Z=V^2=e^{-\frac{C_D\rho_oA}{\beta m}e^{-\beta y}}\left[\frac{2g}{\beta}\sum_{n=1}^{\infty}\frac{\left(\frac{C_D\rho_oA}{\beta m}e^{-\beta y}\right)^n}{n!n}-2gy+const.\right] \tag{5}$$

so that the deceleration becomes, in terms of gravity acceleration,

$$-\frac{\frac{dV}{dt}}{g}=\frac{C_D\rho_oA}{2mg}e^{-\beta y}e^{-\frac{C_D\rho_oA}{\beta m}e^{-\beta y}}\left[\frac{2g}{\beta}\sum_{n=1}^{\infty}\frac{\left(\frac{C_D\rho_oA}{\beta m}e^{-\beta y}\right)^n}{n!n}-2gy+const.\right]-1 \tag{6}$$

As an example, consider the vertical descent of a solid iron sphere having a diameter of 1 foot. For a sphere the drag coefficient may be taken as unity, based on the frontal area for all Mach numbers greater than about 1.4. In equation (2), which describes the variation of density with altitude, the constants should clearly be so chosen as to give accurate values of the density over the range of altitudes for which the deceleration is large. It is seen in figure 1 that for

$$\rho_o=0.0034 \text{ slugs/ft}^3$$

and

$$\beta=\frac{1}{22{,}000}\text{ ft}^{-1}$$

which yields

$$\rho=0.0034\,e^{-\frac{y}{22{,}000}} \tag{7}$$

the calculated density is in good agreement with the NACA standard atmosphere values obtained from references 2 and 3 for the altitude range from 20,000 to 180,000 feet. These relations have been used in calculating the velocity and deceleration of the sphere for various altitudes, assuming

[2] Properly, the analysis should consider those effects resulting from the fact that the earth is a rotating sphere, but since the altitude range for which drag effects are important is less than 1 percent of the radius of the earth, the rectilinear treatment given in this analysis is permissible.

[3] As is well known, this assumption is generally of good accuracy at the high Mach numbers under consideration, at least as long as the total drag is largely pressure drag.

[4] The acceleration of gravity decreases by only 1 percent for every 100,000-foot increase in altitude.